

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 2**

**COURSE NAME: COMPUTER NETWORKS**  
**COURSE CODE: CSA0713**

### **COURSE OUTCOME COVERED**

**CO2:** Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

*Assessment Tool Weightage: 30%*

Annexure A

### **Scenario-Based**

#### ***Code of Conduct:***

*I SHAIK SAMEER(192312356) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

#### ***Signature of the student:***

**SHAIK SAMEER**

## Annexure A

### Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Quality Assurance → /25

Human Resources → /26

Executive Management → /27

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

**2001:0db8:abcd:0000:0000:0000:0000/64**

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the /64 prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

**192.168.2.45**

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

## RUBRICS FOR CALCULATION

| Scenario Based Assessment    |               |                                                                           |                                                    |                                                   |                                                   |
|------------------------------|---------------|---------------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| Criteria                     | Weightage (%) | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                               | Level 2 (Developing)                              | Level 1 (Beginning)                               |
| Understanding of Scenario    | 20%           | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps           | Partial understanding; misses key aspects         | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20%           | Accurately identifies all relevant issues and factors involved            | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                     |
| Application of Concepts      | 25%           | Applies appropriate concepts effectively to address the scenario          | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts                 |
| Decision Making              | 20%           | Makes well-reasoned, logical decisions with strong rationale              | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided     |
| Clarity of Response          | 15%           | Response is clear, structured, and logically presented                    | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand          |

## **Solution:**

### **SCENARIO 1:**

#### **Smart Campus Network Subnetting**

##### **Scenario Understanding**

A Smart Campus has been assigned the IP address block 192.168.100.0/24. Different departments require different subnet sizes based on the expected number of devices. The objective is to allocate IP addresses efficiently using subnetting while minimizing address wastage.

##### **Given Data**

| Department                       | Subnet |
|----------------------------------|--------|
| School of Computing              | /25    |
| School of Electronics            | /26    |
| School of Mechanical Engineering | /27    |

Step 1: Calculate the Subnets

##### **School of Computing**

| Parameter         | Value                           |
|-------------------|---------------------------------|
| Network Address   | 192.168.100.0                   |
| Subnet Mask       | 255.255.255.128                 |
| Broadcast Address | 192.168.100.127                 |
| Host Range        | 192.168.100.1 – 192.168.100.126 |
| Usable Hosts      | 126                             |

##### **School of Electronics**

| Parameter         | Value                             |
|-------------------|-----------------------------------|
| Network Address   | 192.168.100.128                   |
| Subnet Mask       | 255.255.255.192                   |
| Broadcast Address | 192.168.100.191                   |
| Host Range        | 192.168.100.129 – 192.168.100.190 |
| Usable Hosts      | 62                                |

### **School of Mechanical Engineering**

| Parameter         | Value                             |
|-------------------|-----------------------------------|
| Network Address   | 192.168.100.192                   |
| Subnet Mask       | 255.255.255.224                   |
| Broadcast Address | 192.168.100.223                   |
| Host Range        | 192.168.100.193 – 192.168.100.222 |
| Usable Hosts      | 30                                |

### **Remaining Unused Address Range**

| Starting Address | Ending Address  |
|------------------|-----------------|
| 192.168.100.224  | 192.168.100.255 |

### **Observation Table**

| Department  | Network Address    | Host Range          | Broadcast Address | Usable Hosts |
|-------------|--------------------|---------------------|-------------------|--------------|
| Computing   | 192.168.100.0/25   | 192.168.100.1–126   | 192.168.100.127   | 126          |
| Electronics | 192.168.100.128/26 | 192.168.100.129–190 | 192.168.100.191   | 62           |
| Mechanical  | 192.168.100.192/27 | 192.168.100.193–222 | 192.168.100.223   | 30           |

### **Decision**

The subnet allocation efficiently divides the available 192.168.100.0/24 network among all departments. Each department receives sufficient host addresses, and the remaining IP addresses (192.168.100.224–192.168.100.255) are reserved for future expansion.

## **SCENARIO 2:**

### **Multinational Software Company**

#### **Scenario Understanding**

A software company has received the private network 10.10.0.0/16. Each department has different networking requirements. Appropriate subnet masks are assigned to maximize IP utilization while supporting future growth.

### Given Data

| Department           | Subnet |
|----------------------|--------|
| Software Development | /24    |
| Quality Assurance    | /25    |
| Human Resources      | /26    |
| Executive Management | /27    |

### Subnet Allocation

| Department           | Network Address | Host Range              | Broadcast Address | Usable Hosts |
|----------------------|-----------------|-------------------------|-------------------|--------------|
| Software Development | 10.10.0.0/24    | 10.10.0.1–10.10.0.254   | 10.10.0.255       | 254          |
| Quality Assurance    | 10.10.1.0/25    | 10.10.1.1–10.10.1.126   | 10.10.1.127       | 126          |
| Human Resources      | 10.10.1.128/26  | 10.10.1.129–10.10.1.190 | 10.10.1.191       | 62           |
| Executive Management | 10.10.1.192/27  | 10.10.1.193–10.10.1.222 | 10.10.1.223       | 30           |

### Remaining Unused Address Range

| Starting Address | Ending Address |
|------------------|----------------|
| 10.10.1.224      | 10.10.255.255  |

### Decision

Variable Length Subnet Masking (VLSM) efficiently allocates addresses according to department size. The remaining address space can be used for future departments or network expansion.



## SCENARIO 3:

### IPv6 Migration

#### Scenario Understanding

A smart hospital is migrating from IPv4 to IPv6 to support thousands of medical devices and IoT sensors. IPv6 offers a significantly larger address space than IPv4.

#### Given IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000/64

### IPv6 Analysis

| Task                 | Answer                                  |
|----------------------|-----------------------------------------|
| Compressed Address   | 2001:db8:abcd::/64                      |
| Expanded Address     | 2001:0db8:abcd:0000:0000:0000:0000:0000 |
| Network Prefix       | 2001:db8:abcd::/64                      |
| Host Bits            | 64 bits                                 |
| Total Host Addresses | $2^{64} = 18,446,744,073,709,551,616$   |

### Decision

IPv6 provides an enormous address space, making it ideal for hospitals with thousands of connected medical devices, IoT sensors, and future smart healthcare applications.

## SCENARIO 4:

### Routing Table Analysis

#### Scenario Understanding

A router receives a packet destined for 192.168.2.45. The router checks its routing table to identify the matching subnet and forwards the packet using the appropriate interface.

### Routing Table

Network  
192.168.1.0/24  
192.168.2.0/24  
192.168.3.0/24

## Routing Decision

| Question           | Answer                                |
|--------------------|---------------------------------------|
| Destination IP     | 192.168.2.45                          |
| Destination Subnet | 192.168 2.0/24                        |
| Matching Route     | 192.168 2.0/24                        |
| Next-Hop Interface | Interface connected to 192.168 2.0/24 |
| Default Route      | 0.0.0.0/0                             |

## Decision

The router performs a longest-prefix match and forwards the packet through the interface associated with the 192.168.2.0/24 subnet. If no matching route exists, the packet is forwarded using the default route.

## SCENARIO 5:

### Office LAN Configuration

#### Scenario Understanding

A company has two LANs connected through a Layer-3 router. Each LAN requires proper network addressing, host allocation, and gateway configuration to ensure reliable communication.

#### Given Data

| LAN            | Subnet           | Gateway       |
|----------------|------------------|---------------|
| Administration | 192.168.10.0/26  | 192.168.10.1  |
| Finance        | 192.168.10.64/26 | 192.168.10.65 |

#### Network Details

| LAN            | Network Address | Host Range                       | Broadcast Address | Gateway       |
|----------------|-----------------|----------------------------------|-------------------|---------------|
| Administration | 192.168.10.0    | 192.168.10.1–<br>192.168.10.62   | 192.168.10.63     | 192.168.10.1  |
| Finance        | 192.168.10.64   | 192.168.10.65–<br>192.168.10.126 | 192.168.10.127    | 192.168.10.65 |

#### Suggested IP Addresses

### **Administration LAN**

| Computer | IP Address   |
|----------|--------------|
| PC1      | 192.168.10.2 |
| PC2      | 192.168.10.3 |
| PC3      | 192.168.10.4 |

### **Finance LAN**

| Computer | IP Address    |
|----------|---------------|
| PC1      | 192.168.10.66 |
| PC2      | 192.168.10.67 |
| PC3      | 192.168.10.68 |

### **Decision**

The two LANs are correctly subnetted using /26 networks. Each LAN has its own gateway, preventing address conflicts and enabling efficient communication through the Layer-3 router. The design also allows for easy network management and future expansion.